

TRANSCIL · ANDROID ENGINEERING

Architecture & Knowledge Deep Dive

Native Kotlin Android app structure, data flow, KYC/Digio contracts, and a reusable project glossary for TranscilMobileApp .

DATE 2026-07-29	PACKAGE com.example.transcilmobileapp
ARCHITECTURE MVVM · Repository · Retrofit/OkHttp/Gson	STATUS Analysis document · no implementation
CONTRACTS Phase C KYC guide · Identity Digio wiring	AUDIENCE Mobile / platform engineers

Date: 2026-07-29

App: TranscilMobileApp (`com.example.transcilmobileapp`)

Audience: Engineers learning the codebase end-to-end

Status: Analysis dump grounded in code + Phase C / identity specs. No implementation.

Primary contracts read:

1. `docs/superpowers/specs/2026-07-28-phase-c-kyc-digio-integration-guide.md`
2. `docs/superpowers/specs/2026-07-24-android-identity-digio-wiring-design.md`

Locked truths:

- Architecture is **MVVM + Repository + Retrofit/OkHttp/Gson**, with **no DI framework** (manual construction).
- Navigation is **hybrid**: Intent stack for splash/auth/journey/KYC; Navigation Component only inside the home shell.
- **Server is completion authority** for KYC: `GET /v1/me/onboarding` + `OnboardingSync.apply`. Local caches are not truth.
- Digio KYC is Custom Tabs + backend `/v1/kyc/*` (no Digio SDK / vendor secrets on device).
- Tokens live in **EncryptedSharedPreferences** (`TokenStore`). Non-secret KYC drafts may use plain `SharedPreferences` (`KycLocalStorage`).
- Gson maps JSON `snake_case` ↔ Kotlin properties via `@SerializedName`.

Packages covered: `TranscilApp` (root) · `auth` · `splash` · `onboarding` · `journey` · `kyc` · `home` · `rental` · `payment` · `repository` · `core` · `data/network` · `data/local` · `data/model` (+ `auth / onboarding / kyc`) · `demo`

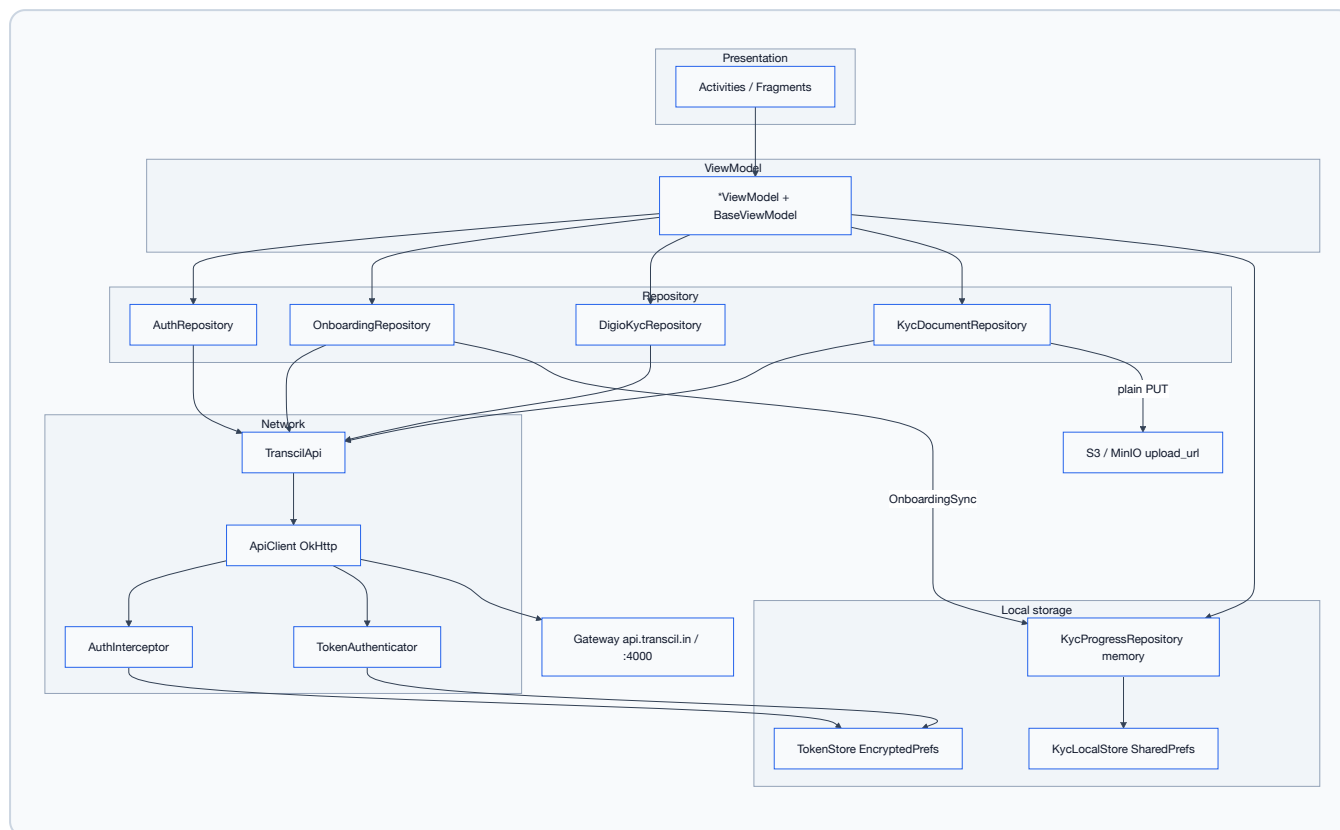
1) Executive map

- **TranscilMobileApp** is a native Kotlin Android rider app (`com.example.transcilmobileapp`) for EV rental / 3PL onboarding and post-KYC home shell.
- **Users:** riders choosing **RENT_EV** (`rider`) or **THREE_PL** (`3pl`) after phone OTP login.
- **Spine:** MVVM + thin Repositories + Retrofit/OkHttp/Gson; **no DI** — `ApiClient` singleton + default ctor injection.
- **Identity/KYC** talks to the Transcil **gateway** (`BuildConfig.BASE_URL` , prod `https://api.transcil.in/` ; local via emulator `10.0.2.2:4000`). Digio runs in **Custom Tabs**; app never holds Digio secrets.
- **Completion authority:** `GET /v1/me/onboarding` → `OnboardingSync.apply`. `KycProgressRepository` / `KycLocalStorage` are cache/draft only.
- **Package ownership:** `auth / splash / onboarding / journey` = entry + session; `kyc` + `repository` = verification funnel; `home` = Nav-shell dashboard; `rental / payment` = UI shells (mostly local catalog); `data.*` = network/storage/DTOs; `core` = shared bases/enums/nav helpers.
- **Navigation hybrid:** Intent stack through splash→auth→journey→KYC; Navigation Component only inside `HomeDashboardActivity`.

- **Tokens** in `EncryptedSharedPreferences` (`TokenStore`); bearer via `AuthInterceptor` ; 401 refresh via `TokenAuthenticator` .
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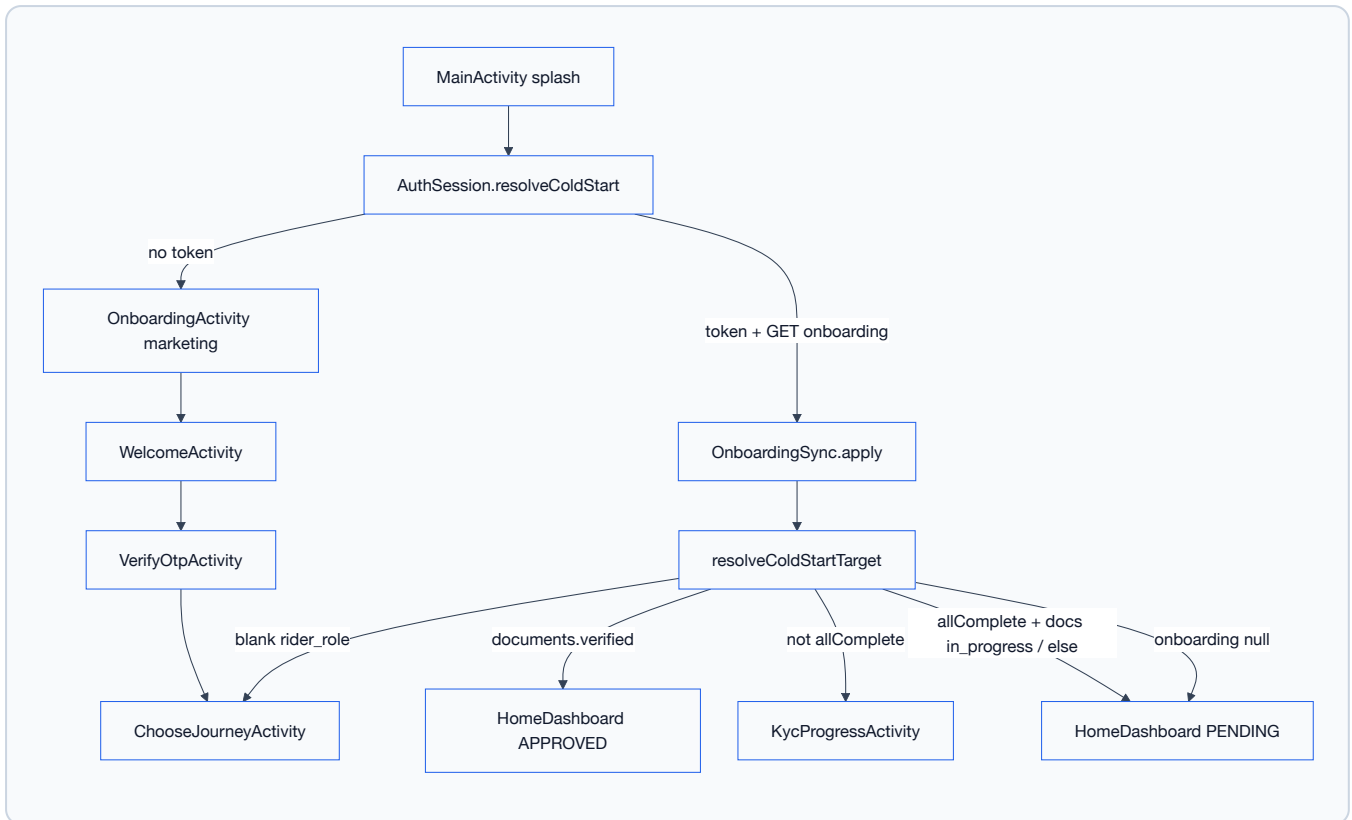
2) Mermaid diagrams

A. Package / layer diagram



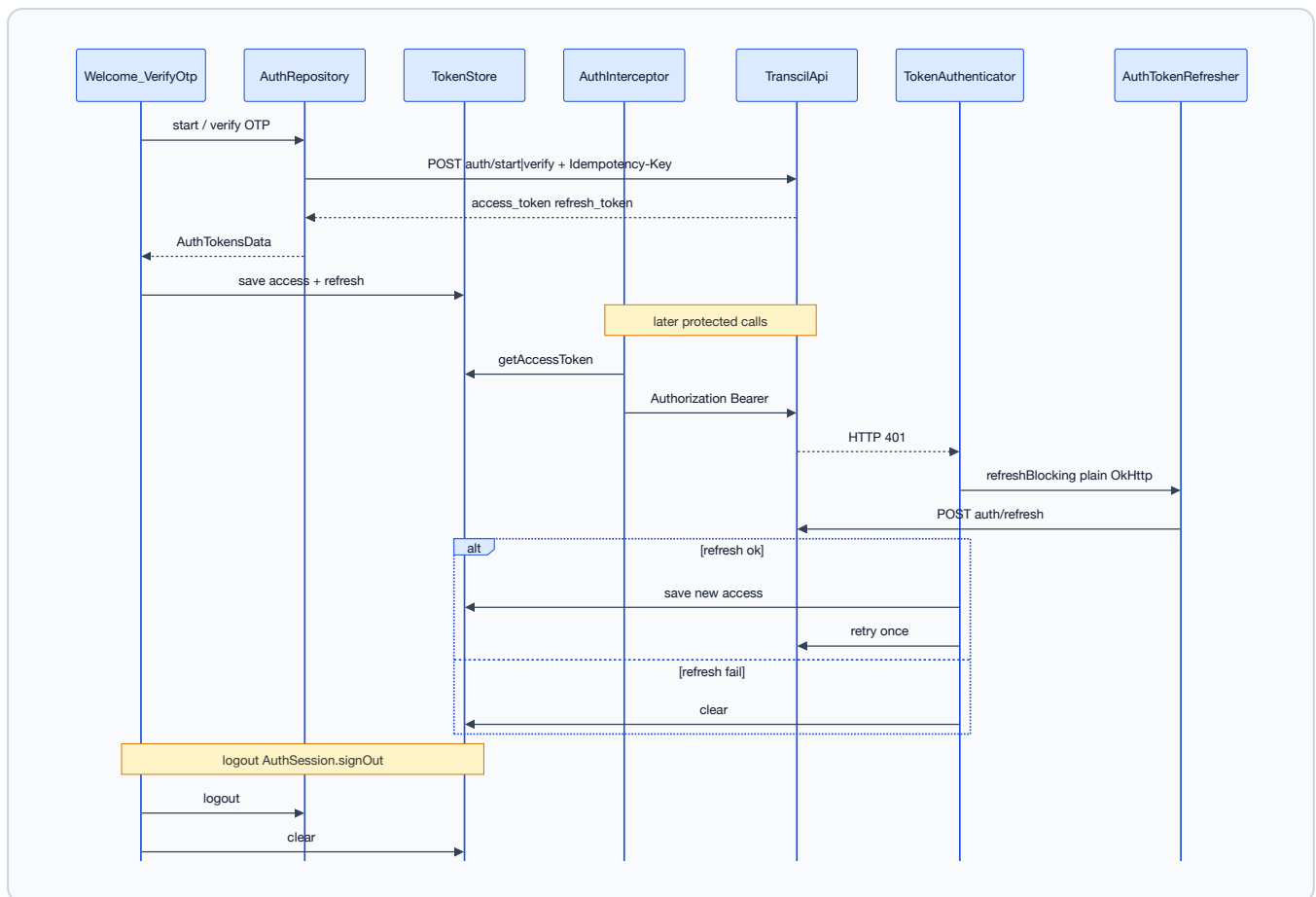
Why this shape: Thin repos keep Activities off Retrofit; OkHttp owns auth headers/refresh so ViewModels stay dumb. No DI keeps the young codebase small. Digio/S3 stay outside the JSON Retrofit client so vendor secrets and multipart never enter the app envelope.

B. Cold-start + auth routing



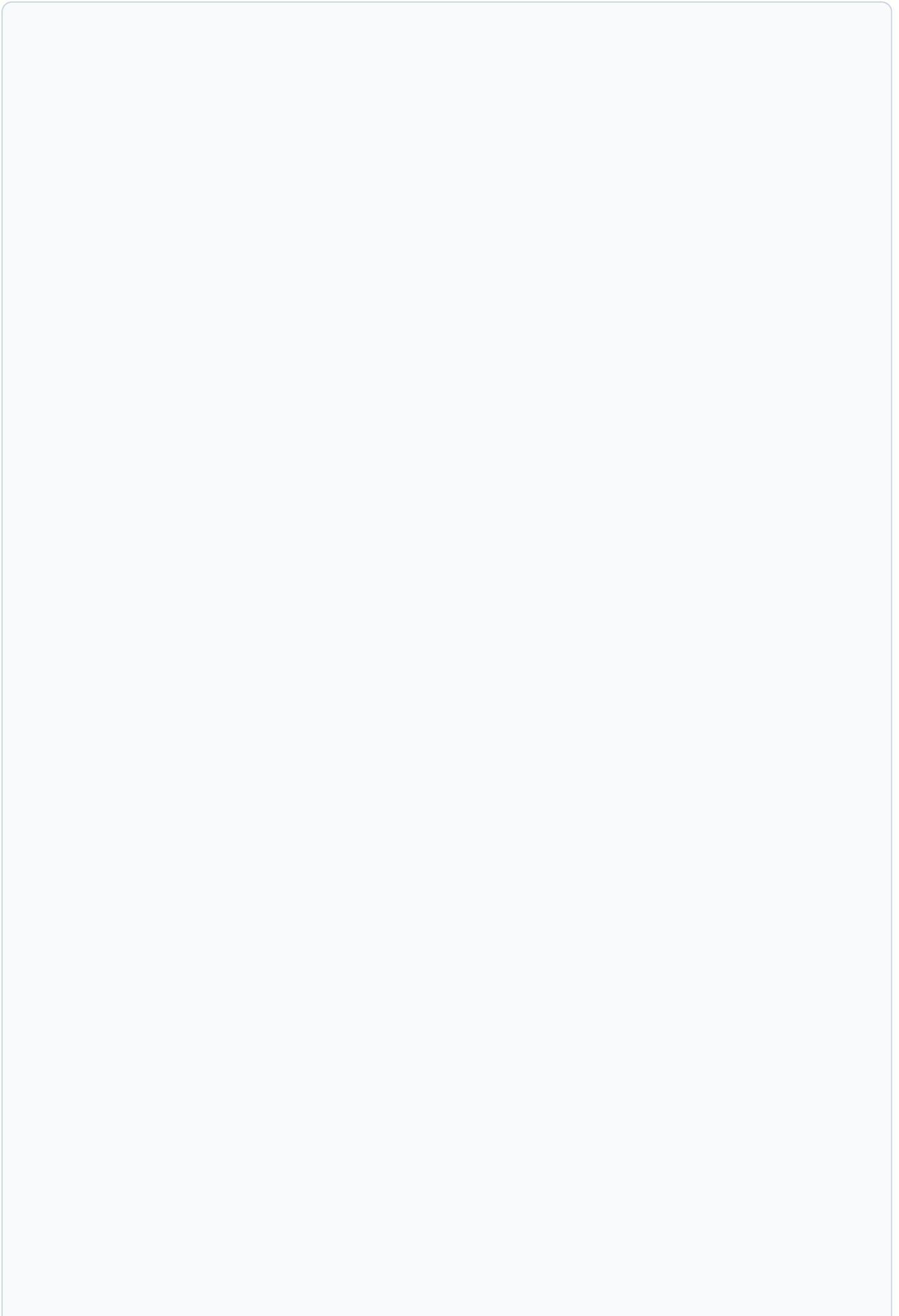
Why: Cold start must survive process death using encrypted tokens + server onboarding, not in-memory KYC. Marketing onboarding is skipped when a session exists. `CLEAR_TASK` intents avoid stacking splash under home/KYC.

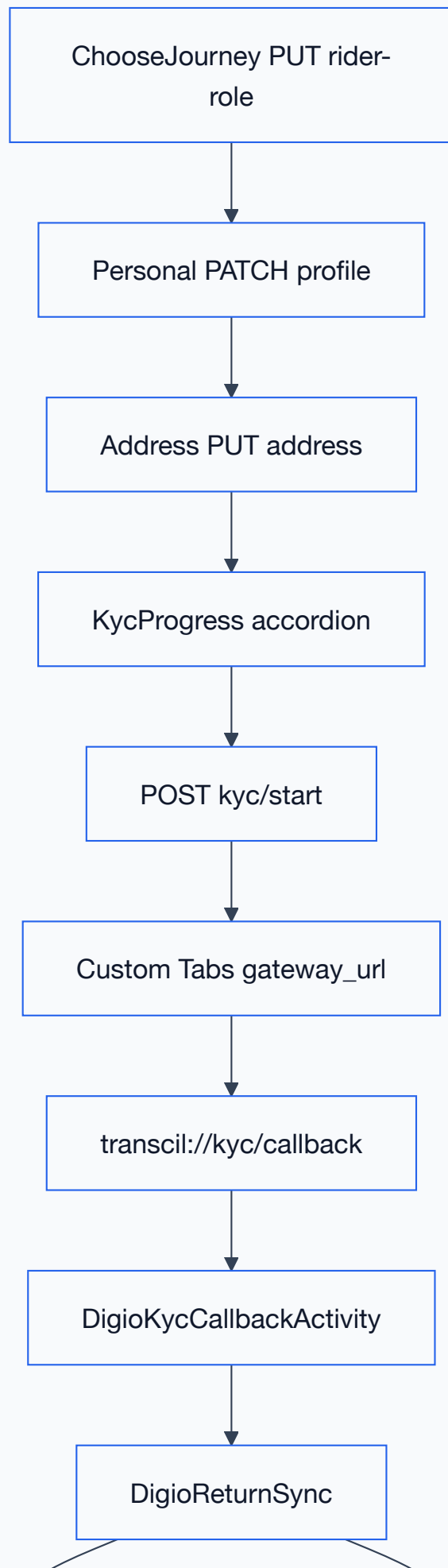
C. Auth token lifecycle

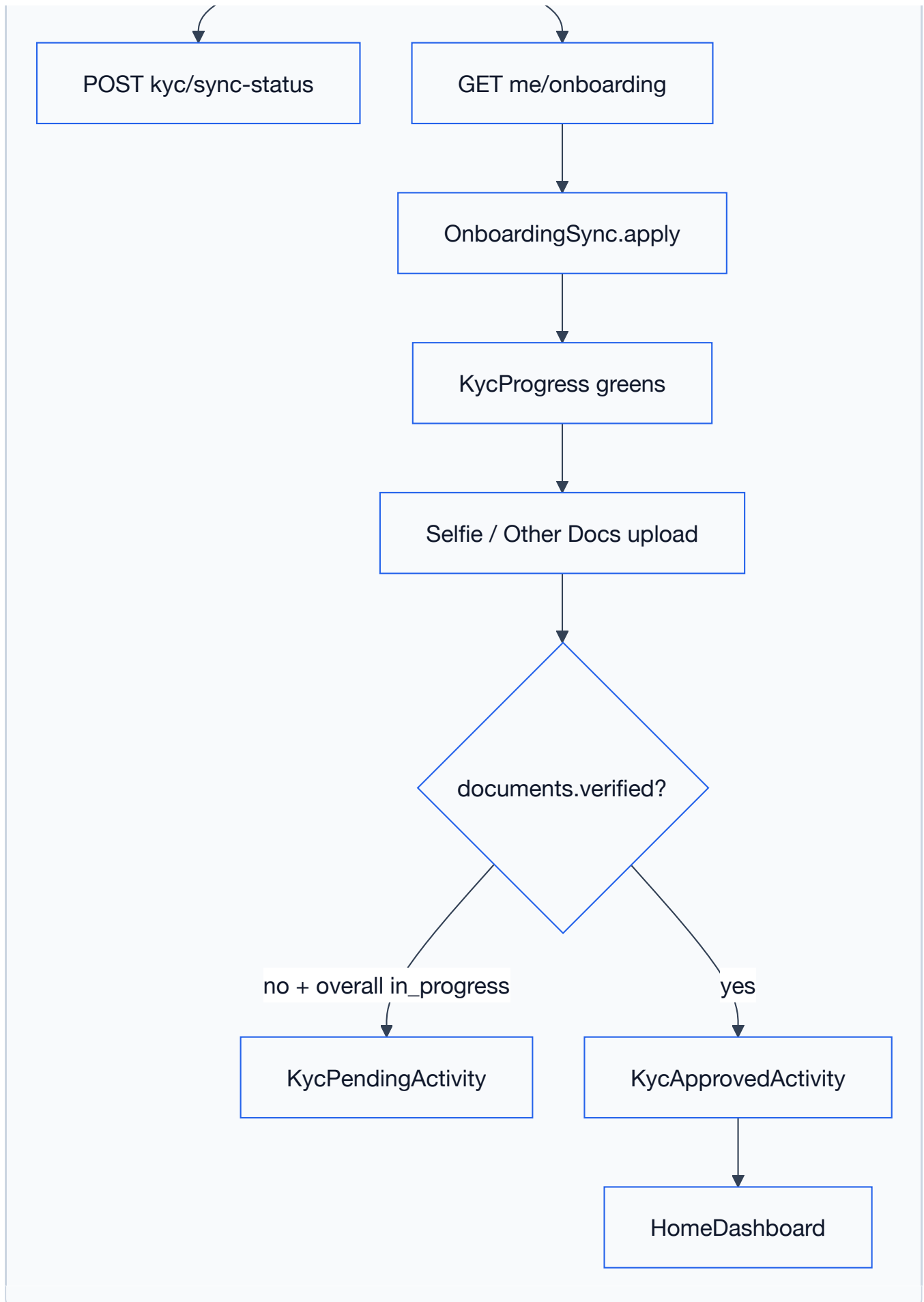


Why: Cognito-backed OTP without Amplify; encrypted disk so restart keeps session; separate plain client for refresh avoids 401 loops; single-flight lock handles concurrent 401s.

D. KYC happy path (Phase C)

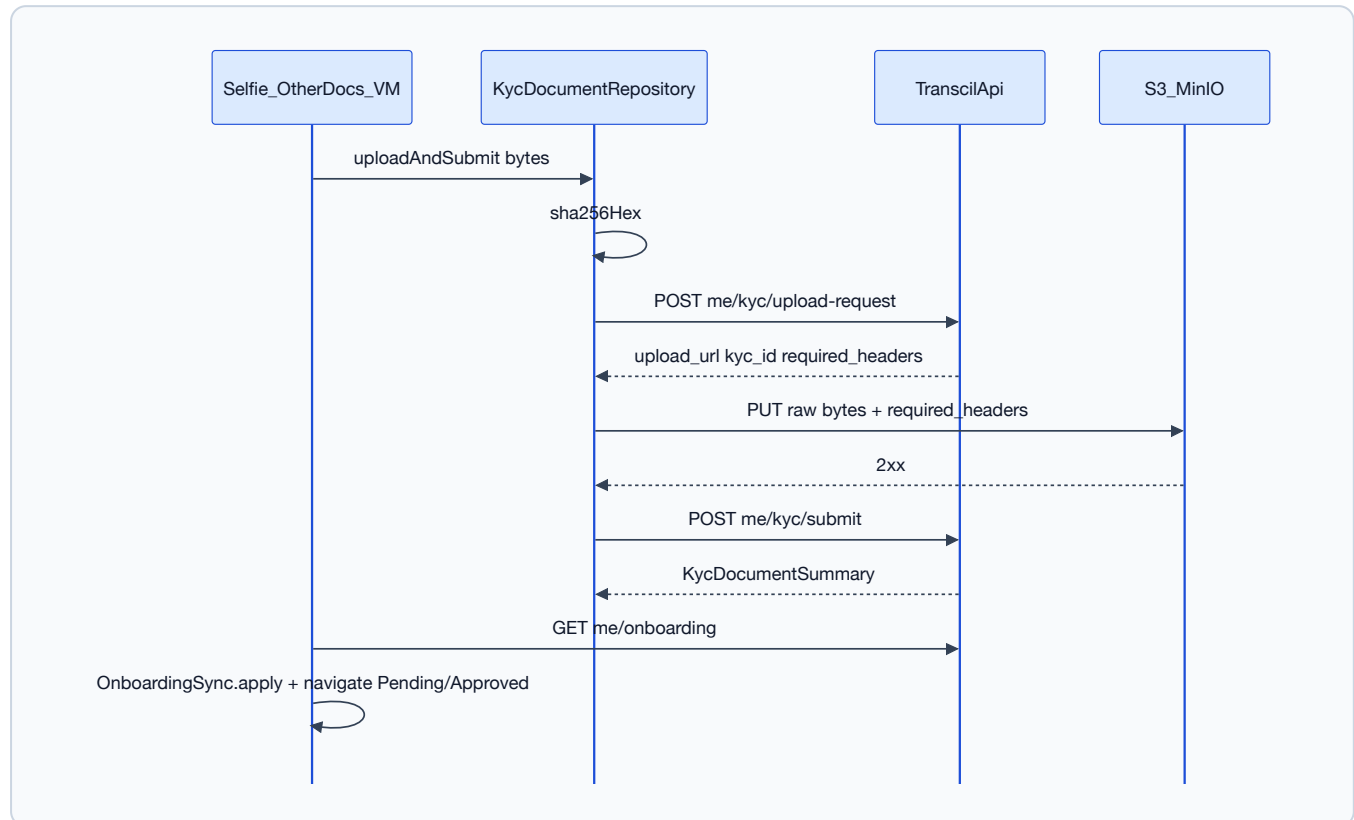






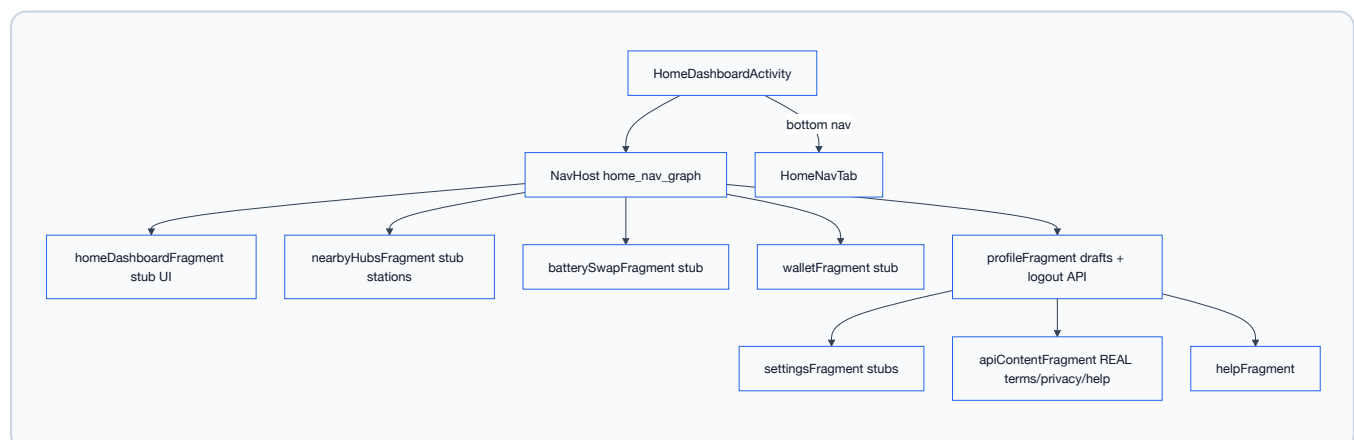
Why: Digio-only for Aadhaar+Bank (Approach A) avoids dual-path drift. Server onboarding after sync prevents fake greens. Deep link matches Digio `redirect_url` so return works after Custom Tabs process handoff.

E. Document upload pipeline



Why: Presign + plain PUT matches identity contract (not multipart through gateway). SHA-256 + required headers satisfy S3 signing; submit is separate so failed PUTs don't create false KYC rows.

F. Home shell



Why: One Activity + Nav graph for tab restore/state; Intent-based KYC stays outside so identity funnel can clear the task. Content pages reuse public GETs already on TranscilApi ; ops features (wallet/swap/hubs) stay stubbed until later phases.

3) File / class catalog by package

Root — TranscilApp.kt

Item	Role / why / spine / deps
TranscilApp	Application ; inits TokenStore + KycLocalStore before Activities. Depends on local stores; depended on by manifest android:name .

auth

File	Role	Why	Key spine	Depends / depended
AuthSession	Cold-start + logout glue	Server onboarding → destination without Activity logic	ColdStartTarget , resolveColdStart , resolveColdStartTarget , signOut , openSignedOut	TokenStore, OnboardingRepository, OnboardingSync, KycProgressRepository; used by MainActivity, Profile
WelcomeActivity / WelcomeViewModel	Phone entry → OTP start	Thin UI over AuthRepository	onSendOtpClicked	AuthRepository → VerifyOtp
VerifyOtpActivity / VerifyOtpViewModel	OTP verify → persist tokens	Session boundary	onVerifyClicked → TokenStore.save	AuthRepository, TokenStore → ChooseJourney

splash

File	Role	Why	Spine	Deps
MainActivity	Branded splash then route	Single launcher entry	navigateAfterSplash → AuthSession.resolveColdStart	AuthSession, BaseActivity

onboarding (marketing — not /me/onboarding)

File	Role	Why	Spine	Deps
OnboardingActivity / OnboardingViewModel	3-slide marketing carousel	Pre-auth product story	pages → navigateToWelcome	→ WelcomeActivity

journey

File	Role	Why	Spine	Deps
ChooseJourneyActivity / ChooseJourneyViewModel	RENT_EV vs THREE_PL	Server role via PUT /v1/me/rider-role	setRiderRole , startJourney	OnboardingRepository, KycProgressRepository → CreatePersonalAccount

kyc (largest)

File / object	Role	Why
KycProgressActivity + KycProgressViewModel	Accordion hub: refresh onboarding, Digio launch, inline forms, other-docs upload, status redirect	Single place for progress UX
KycProgressRepository	In-memory drafts + step UI state	Fast accordion; overwritten by OnboardingSync
OnboardingSync	Maps OnboardingData → local drafts/completion	Server = truth
DigioReturnSync	sync-status then onboarding refresh	Phase C correction vs mark-local-complete
DigioKycCallbackActivity	Deep-link receiver	transcil://kyc/callback
DigioLauncher	Custom Tabs open	No Digio SDK
CreatePersonalAccount*	Personal details	PATCH profile
AddressDetails*	Address	PUT address + states/cities
AadhaarVerification* / AadhaarOtp*	Legacy/direct Aadhaar UI	Phase C: Digio happy path; OTP dormant
BankDetails*	Bank form + local IFSC draft	Digio primary; local-only bank completion still in repo (ponytail)
PanVerification*	3PL PAN	verifyPan
SelfieVerification*	Selfie capture → upload pipeline	Docs gate
OtherDocsCatalog / validators	Doc type labels + validation	From onboarding options.doc_types
KycPendingActivity / KycApprovedActivity	Review / approved gates	Driven by documents.*
KycFlowNavigator / KycStepModels	Step intents + catalog	Local step order by journey
KycStepCatalog	Hard-coded step lists per journey	UI order; completion still from server keys

Depends on: repositories, TokenStore indirectly, KycLocalStore. Depended on by: AuthSession, home Profile (documents browse).

home

File	Role	API?
HomeDashboardActivity	Shell + bottom nav	Nav only
HomeDashboardFragment / ViewModel	Home tab	Stub name/id from drafts
NearbyHubs* / BatterySwap* / Wallet*	Ops tabs	In-memory stubs

File	Role	API?
Profile*	Profile + logout	Logout → AuthSession; display from drafts
ApiContent* / Help*	Terms/privacy/help	Real TranscilApi GETs
Settings*	Settings menu	Mostly stubs
RentalCatalog / HomeModels	Shared demo catalog	Local prices for rental/payment

rental / payment

Package	Role	Why
VehiclesActivity , RentalPlans*	Catalog UI	Local RentalCatalog ; not booking API
PaymentActivity / PaymentViewModel	Fake pay → autopay → success steps	UI prototype; no payment API

repository

Class	Owns	Maps
AuthRepository	/v1/auth/* , toE164	Envelope errors → Result
OnboardingRepository	journey, role, profile, address, onboarding, reference, PAN, states/cities	UI DOB/gender ↔ API
DigioKycRepository	`v1/kyc/start	sync`
KycDocumentRepository	upload-request → S3 PUT → submit	SHA-256, required headers
DemoRepository	Demo ping	Early networking sample

core

Symbol	Role
BaseActivity	ViewBinding inflate + window insets
BaseViewModel	isLoading / errorMessage LiveData
JourneyType , Gender , KycStatus	Shared enums
KycNavigator	Pending/Approved/Home Intents
NavExtras , OtpInput , FeedbackUi , UiFormHelpers , SegmentedStepper	Cross-cutting UI helpers

data/network

Class	Role
ApiClient	Retrofit + OkHttp (interceptor, authenticator, debug logging)
TranscilApi	All gateway endpoints

Class	Role
AuthInterceptor	Bearer attach
TokenAuthenticator + AuthTokenRefresher	401 → refresh once
DemoApi	Demo endpoints

data/local

Class	Role
TokenStore	Encrypted access/refresh + memory cache
KycLocalStore	Plain prefs bank draft (process death)

data/model

Area	Role
ApiResponse / ApiMeta / ApiError	Envelope
auth/AuthDtos	OTP/token bodies
onboarding/OnboardingDtos	Onboarding + profile/address/journey
kyc/DigioDtos , KycDocumentDtos	Digio + upload
PublicContentDtos	Terms/privacy/help

demo

File	Role
DemoViewModel	Early Retrofit smoke sample via DemoRepository

4) Concept deep-dives

1. SharedPreferences vs EncryptedSharedPreferences

- **TokenStore** (`data/local/TokenStore.kt`): `EncryptedSharedPreferences` + `MasterKey AES256`; keys `access_token` / `refresh_token`; in-memory mirror for `OkHttp` threads. **Why encrypt:** bearer/refresh are session secrets; disk readable otherwise.
- **KycLocalStore** : plain `SharedPreferences` (`transcil_kyc_local`) for bank draft fields + completed subtitle. **Why not encrypt:** non-secret UX draft / IFSC path survival across process death; Phase C still forbids persisting full Aadhaar/PAN/OTP — bank account in plain prefs is a known trade-off for the local IFSC path (`ponytail` in `KycProgressRepository`).
- **Logout** (`AuthSession.signOut`): best-effort `authRepository.logout()` , then `TokenStore.clear()` , `KycProgressRepository.clearAuthLocal()` (resets memory + `KycLocalStore.clear()`). Lands on `WelcomeActivity` via `openSignedOut` (not marketing onboarding).

2. @SerializedName

Gateway identity JSON is **snake_case**; Kotlin idioms are **camelCase**. Gson needs explicit names (or a field-naming policy — this app uses annotations).

Examples:

- Auth: `phone_e164` → `phoneE164` ; `access_token` → `accessToken` (`AuthDtos.kt`)
- Onboarding: `rider_role` , `overall_percent` , `completed_at` , `edit_endpoint` (`OnboardingDtos.kt`)
- Digio: `gateway_url` , `redirect_url` , `customer_name` (`DigioDtos.kt`)

Alternate names: `ApiMeta.requestId` uses `@SerializedName(value = "requestId", alternate = ["request_id"])` so older camelCase content responses and identity snake_case both parse. Same pattern on `sort_order` / `state_code` in reference DTOs.

3. MVVM here

- **BaseActivity** : inflates `ViewBinding`, safe insets — Activities observe `LiveData` and call VM methods.
- **BaseViewModel** : shared loading/error; feature VMs add navigation `LiveData` (`VerifyOtpViewModel.navigateToChooseJourney`).
- Work runs in `viewModelScope.launch` ; UI must not call `Retrofit` — pattern is VM → Repository → `ApiClient.transcilApi`. Exception: `ApiContentViewModel` calls `ApiClient.transcilApi` directly (small YAGNI for public HTML).

4. Repository layer

Layer	Owns
Validators (<code>*Validator.kt</code>)	Pure field rules (regex, lengths, consent)
ViewModels	UX orchestration, when to navigate, toast, loading

Layer	Owns
Repositories	E.164/ +91 , DOB format, Idempotency-Key UUIDs, envelope <code>error</code> → <code>Result.failure</code> , Digio name validation, S3 PUT

5. Idempotency-Key, AuthInterceptor, TokenAuthenticator

- **Idempotency-Key:** UUID per logical POST/PUT (auth, Digio, profile, uploads). Prevents duplicate Digio sessions / KYC rows on retries. Reuse only when retrying the same action after timeout (contract).
- **AuthInterceptor:** attaches Bearer when token present; public GETs work pre-login.
- **TokenAuthenticator:** on 401, skip refresh path, cap one retry, single-flight; uses **plain** `AuthTokenRefresher` `OkHttp` (no authenticator) to avoid loops; clears tokens on hard failure so next cold start → marketing/onboarding path without zombie Bearer.

6. OnboardingSync vs KycProgressRepository

- **Source of truth:** server `OnboardingData` via `OnboardingRepository.getOnboarding()` .
- **OnboardingSync.apply** : maps `steps[].key/status/fields` into drafts + `syncStepStatuses` ; restores bank from `KycLocalStore` first so local-only bank isn't wiped.
- **KycProgressRepository** : session drafts, accordion `uiSteps()` , sequencing `canOpen` ; **not** authority for “step complete” after Digio (Phase C). `DigioReturnSync` always refreshes onboarding rather than `markCompleted` from Digio alone.

Doc vs code: Phase B design (2026-07-24) said Digio sync may mark Aadhaar/Bank locally. Phase C guide (2026-07-28) + `DigioReturnSync` are **newer** — onboarding-only completion.

7. Intent vs Navigation Component

- **Identity funnel** = linear, rare back-stack restore needs, deep links, task clears → Activities + Intents.
- **Home** = multi-tab with nested Settings/Help → `home_nav_graph.xml` + bottom nav state restore. Keeps Navigation dependency scoped to the shell.

8. Process death / resume

Survives	Dies
Encrypted tokens (<code>TokenStore</code>)	In-memory drafts (personal/address/...) unless rehydrated from onboarding
<code>KycLocalStore</code> bank draft	Accordion expand state, pending selfie bytes in VM
Server onboarding (refetch on cold start / <code>KycProgressViewModel.refresh</code>)	Digio Custom Tabs mid-flow (return via deep link)

Cold start with token: `MainActivity` → `resolveColdStart` → GET onboarding → `OnboardingSync.apply` → correct Activity.

5) End-to-end runtime walkthroughs

1. First install → session saved

MainActivity splash → AuthSession.resolveColdStart (no token) → OnboardingActivity → Get Started → WelcomeActivity → WelcomeViewModel.onSendOtpClicked → AuthRepository.start (+91...) → VerifyOtpActivity → VerifyOtpViewModel.onVerifyClicked → AuthRepository.verify → TokenStore.save → ChooseJourneyActivity.

2. Choose journey → KYC accordion

ChooseJourneyViewModel.onContinueClicked → setRiderRole → KycProgressRepository.startJourney → CreatePersonalAccount* (patchProfile) → address → KycProgressActivity .Accordion from KycStepCatalog + server statuses via refresh → OnboardingSync.apply .

3. Digio return → greens

Aadhaar/Bank CTA → DigioKycRepository.start → DigioLauncher.open(gateway_url) → Digio → deep link → DigioKycCallbackActivity → DigioReturnSync.applyAfterReturn (sync-status + getOnboarding + OnboardingSync.apply) → KycProgressActivity with updated completes / in_progress.

4. Selfie / other docs → Home

SelfieVerificationViewModel.onContinue or other-docs path in KycProgressViewModel → KycDocumentRepository.uploadAndSubmit → GET onboarding → if not verified → KycPendingActivity / PENDING home; if documents.verified → KycApprovedActivity → KycNavigator.openHomeDashboard .

5. Cold start with valid token

Splash → shouldRestoreSession true → getOnboarding → OnboardingSync.apply → resolveColdStartTarget : blank role → ChooseJourney; verified → Home APPROVED; incomplete steps → KycProgress; else Home PENDING.

6. Mid-session 401

Protected call 401 → TokenAuthenticator.authenticate → AuthTokenRefresher.refreshBlocking → success: TokenStore.save + retry; fail: TokenStore.clear (next navigation/cold start treats as logged out). UI does not implement refresh itself.

7. Logout

Profile → ProfileViewModel.onLogout → AuthSession.signOut (API logout best-effort, clear tokens + KYC local) → AuthSession.openSignedOut → WelcomeActivity with NEW_TASK|CLEAR_TASK .

6) What is NOT backed by API yet

- Home dashboard quick actions / stub Transcil ID (`HomeDashboardViewModel`)
- Nearby hubs, battery swap, wallet (dedicated stub repositories)
- Profile edit, many settings toggles, support stubs
- Rental vehicles/plans + Payment flow (`RentalCatalog` + local state machine)
- Direct Aadhaar OTP happy path / `POST /v1/me/bank-account` (Phase C out of scope)
- Booking, real wallet money, hub maps (Phase C doc: out of scope)

Is API-backed: auth, Digio KYC, onboarding/profile/address/reference/PAN, document upload, public content (terms/privacy/help/settings GETs).

7) Project term cheatsheet

Term	Definition here	Where	Why it matters
Transcil	Rider EV rental / 3PL platform; this is the Android client	App + <code>api.transcil.in</code>	Product name / API host
KYC	Know-your-customer verification funnel	<code>kyc/*</code> , <code>/v1/kyc/*</code> , <code>/v1/me/kyc/*</code>	Gate to Home
Digio	Hosted Aadhaar+Bank KYC vendor via backend	<code>DigioKycRepository</code> , Custom Tabs	No Digio SDK/secrets on device
Aadhaar	National ID step; Digio-hosted	Step key <code>aadhaar</code> , <code>KycStep.AADHAAR</code>	Identity verify
PAN	Tax ID; 3PL one-shot verify	<code>verifyPan</code> , <code>KycStep.PAN</code>	3PL checklist
IFSC	Bank branch code in local bank draft	<code>BankDraft.ifsc</code> , <code>KycLocalStore</code>	Local bank form UX
OTP	SMS one-time password for login (and dormant Aadhaar OTP UI)	<code>AuthRepository</code> , <code>VerifyOtp</code>	Auth entry
e164	+91 + 10 digits phone form	<code>AuthRepository.toE164</code> , reference <code>mobile_e164</code>	API phone format
RENT_EV / rider	Rental journey / API role <code>rider</code>	<code>JourneyType.RENT_EV</code> , <code>toRiderRole</code>	Step set + product
THREE_PL / 3pl	Logistics journey / API role <code>3pl</code>	<code>JourneyType.THREE_PL</code>	Includes PAN; different catalog
Onboarding (marketing)	Pre-login carousel	<code>onboarding/OnboardingActivity</code>	Not the API resource
Onboarding (API)	<code>GET /v1/me/onboarding</code> progress	<code>OnboardingData</code> , <code>OnboardingRepository</code>	Completion authority
OnboardingSync	Apply API onboarding → local drafts/status	<code>kyc/OnboardingSync.kt</code>	Prevents fake greens
DigioReturnSync	Post-deep-link sync + onboarding refresh	<code>kyc/DigioReturnSync.kt</code>	Digio return contract
KycProgressRepository	In-memory KYC drafts + UI step state	<code>kyc/KycProgressRepository.kt</code>	Cache/draft, not truth
KycLocalStore	Plain prefs bank draft	<code>data/local/KycLocalStore.kt</code>	Survives process death

Term	Definition here	Where	Why it matters
TokenStore	Encrypted access/refresh tokens	<code>data/local/TokenStore.kt</code>	Session persistence
EncryptedSharedPreferences	AES-backed prefs for secrets	TokenStore via <code>security-crypto</code>	Token at rest
SharedPreferences	Plain prefs	KycLocalStore	Non-secret drafts
ApiClient	Retrofit/OkHttp singleton	<code>data/network/ApiClient.kt</code>	All API access
TranscilApi	Retrofit interface of gateway routes	<code>data/network/TranscilApi.kt</code>	Contract surface
ApiResponse	<code>{ data, meta, error }</code> envelope	<code>data/model/ApiResponse.kt</code>	Uniform parsing
@SerializedName	Gson JSON↔Kotlin name bridge	DTOs under <code>data/model</code>	snake_case APIs
Idempotency-Key	Per-action UUID header on mutating calls	Repos + <code>TranscilApi @Header</code>	Safe retries
AuthInterceptor	Adds <code>Authorization: Bearer</code>	<code>AuthInterceptor.kt</code>	Authenticated calls
TokenAuthenticator	401 → refresh once → retry	<code>TokenAuthenticator.kt</code>	Session continuity
AuthSession	Cold-start routing + logout	<code>auth/AuthSession.kt</code>	Entry glue
ColdStartTarget	Sealed destinations after splash	<code>AuthSession.kt</code>	Testable routing
Custom Tabs	Chrome custom tab for Digio URL	<code>DigioLauncher</code> , <code>androidx.browser</code>	Safer than WebView
gateway_url	Digio start response URL to open	<code>DigioStartData.gatewayUrl</code>	Hosted KYC entry
transcil://kyc/callback	Digio redirect deep link	Manifest + <code>DIGIO_REDIRECT_URL</code>	Return to app
documents.overall	Doc review aggregate (pending / in_progress / ...)	<code>OnboardingDocumentsStatus.overall</code>	Pending screen
documents.verified	Boolean unlock for approved Home	<code>OnboardingDocumentsStatus.verified</code>	Final gate
presign / S3 PUT	upload-request then binary PUT	<code>KycDocumentRepository</code>	Doc upload contract
Other Docs	Non-selfie KYC docs (voter/DL/PAN file)	<code>OTHER_DOCS</code> , <code>OtherDocsCatalog</code>	Rental checklist

Term	Definition here	Where	Why it matters
Reference	Emergency contact for rider	PUT /v1/me/reference , KycStep.REFERENCE	Rental required step
Phase A/B/C	Delivery slices: auth wiring → Digio → full onboarding truth	Specs under docs/superpowers/specs/	Scope discipline
ViewBinding / DataBinding	Generated bindings; DataBinding enabled in Gradle	BaseActivity , build.gradle.kts	UI wiring
MVVM	UI ↔ ViewModel ↔ Repository	Throughout	Separation
Repository pattern	Thin API + mapping layer	repository/	Testable boundary
LiveData	UI observation	ViewModels	Lifecycle-safe
viewModelScope	Coroutine scope tied to VM	ViewModels	Cancel on clear
BuildConfig.BASE_URL	Retrofit base; default prod	app/build.gradle.kts	Environment switch
local gateway	Docker nginx :4000 ; emulator http://10.0.2.2:4000/	Phase C guide; debug cleartext manifest	Local integration
KycNavigator	Pending/Approved/Home intents	core/KycNavigator.kt	Post-submit routing
KycStepCatalog	Local step order per journey	kyc/KycStepModels.kt	Accordion composition
AuthTokenRefresher	Blocking refresh without authenticator	AuthTokenRefresher.kt	Breaks refresh loops
client_action	Server hint for UI (stay_on_screen , etc.)	ApiError.clientAction	Error UX
Browse-only KYC	Profile → progress without status redirect	EXTRA_BROWSE_ONLY	Approved riders can view docs

Open questions / gaps

- **KycStepCatalog vs server `steps[]` order:** completion maps by key, but accordion order is still hard-coded (and 3PL order differs from Phase C's "typical" list). Unclear if product will switch to server order only.
- **Local bank completion** (`markCompletedLocalOnly` + `KycLocalStore`) still conflicts with Phase C "don't locally complete BANK" — intentional interim or leftover?
- **Cold start when onboarding fetch fails** (`onboarding == null` → Home PENDING): may strand users; error UX not obvious from code.
- **Release cleartext off** (`usesCleartextTraffic=false`) while prod URL is HTTPS — fine for prod; local HTTP needs debug build or `BASE_URL` override (not in checked-in `defaultConfig`).
- **Profile/home still show stub rider IDs** even after API profile exists — `getProfile` not wired into Profile bind yet.

Generated from `docs/superpowers/specs/2026-07-29-transcil-android-architecture-deep-dive.md` . Mermaid diagrams are rendered as vector graphics for device-independent reading.